

SPRING DATA JPA + ORM (EXPLANATION + OUTPUT)

1. WHAT YOU ARE LEARNING

You are building a Spring Boot project that demonstrates:

Query Methods (no SQL needed)

ORM Relationships

- ManyToOne
- OneToMany
- ManyToMany

Fetching data using JPA repositories

2. QUERY METHODS THEORY

Spring Data JPA automatically converts method names into SQL.

PROJECT STRUCTURE

```
orm-learn
|
├── model
|   ├── Country.java
|   ├── Stock.java
|   ├── Employee.java
|   ├── Department.java
|   └── Skill.java
|
├── repository
|   ├── CountryRepository.java
|   ├── StockRepository.java
|   ├── EmployeeRepository.java
|   ├── DepartmentRepository.java
|   └── SkillRepository.java
|
├── service
|   ├── EmployeeService.java
|   ├── DepartmentService.java
|   └── SkillService.java
|
└── OrmLearnApplication.java
```

HANDS-ON 1: COUNTRY QUERY METHODS

THEORY

Spring Data JPA provides Query Methods, where SQL is automatically generated from method names.

No SQL needed

Method name defines query logic

Supports filtering, sorting, searching

Repository Code

```
public interface CountryRepository extends JpaRepository<Country, String> {  
  
    List<Country> findByNameContaining(String name);  
  
    List<Country> findByNameContainingOrderByNameAsc(String name);  
  
    List<Country> findByNameStartingWith(String letter);  
}
```

Test Code

```
List<Country> list = countryRepository.findByNameContaining("ou");  
System.out.println(list);
```

```
List<Country> sorted =  
countryRepository.findByNameContainingOrderByNameAsc("ou");  
System.out.println(sorted);
```

```
List<Country> zList = countryRepository.findByNameStartingWith("Z");  
System.out.println(zList);
```

OUTPUT

```
BV Bouvet Island  
DJ Djibouti  
GP Guadeloupe
```

GS South Georgia and the South Sandwich Islands
LU Luxembourg
SS South Sudan
TF French Southern Territories
UM United States Minor Outlying Islands
ZA South Africa

Sorted result:

BV Bouvet Island
DJ Djibouti
GP Guadeloupe
LU Luxembourg
TF French Southern Territories
ZA South Africa
GS South Georgia and the South Sandwich Islands
SS South Sudan
UM United States Minor Outlying Islands

✓ Starting letter:

ZM Zambia
ZW Zimbabwe

HANDS-ON 2: STOCK QUERY METHODS

THEORY

Spring Data JPA supports:

Range queries → Between dates

Comparison → GreaterThan, LessThan

Sorting → Top, OrderBy

Filtering by multiple conditions

Repository Code

```
public interface StockRepository extends JpaRepository<Stock, Integer> {
```

```
    List<Stock> findByStCodeAndStDateBetween(String code, Date start, Date
```

```
end);
```

```
List<Stock> findByStCodeAndStCloseGreaterThan(String code, Double price);
```

```
List<Stock> findTop3ByOrderByStVolumeDesc();
```

```
List<Stock> findTop3ByStCodeOrderByStCloseAsc(String code);  
}
```

Test Code

```
stockRepository.findByStCodeAndStDateBetween(  
    "FB",  
    Date.valueOf("2019-09-01"),  
    Date.valueOf("2019-09-30")  
);
```

```
stockRepository.findByStCodeAndStCloseGreaterThan("GOOGL", 1250);
```

```
stockRepository.findTop3ByOrderByStVolumeDesc();
```

```
stockRepository.findTop3ByStCodeOrderByStCloseAsc("NFLX");
```

OUTPUT

Facebook (September 2019)

```
FB 2019-09-03 184.00 182.39 9779400  
FB 2019-09-04 184.65 187.14 11308000  
FB 2019-09-05 188.53 190.90 13876700  
...  
FB 2019-09-27 180.49 177.10 14656200
```

Fetch all Facebook stock records in given date range.

Google (Price > 1250)

```
GOOGL 2019-04-22 1236.67 1253.76 954200  
GOOGL 2019-04-23 1256.64 1270.59 1593400  
GOOGL 2019-04-24 1270.59 1260.05 1169800  
GOOGL 2019-04-25 1270.30 1267.34 1567200
```

GOOGL 2019-04-26 1273.38 1277.42 1361400
GOOGL 2019-04-29 1280.51 1296.20 3618400
GOOGL 2019-10-17 1251.40 1252.80 1047900

Filter stocks where closing price > 1250

Top 3 Volume Stocks

FB 2019-01-31 165.60 166.69 77233600
FB 2018-10-31 155.00 151.79 60101300
FB 2018-12-19 141.21 133.24 57404900

Meaning: Highest traded volume stocks

Netflix Lowest 3

NFLX 2018-12-24 242.00 233.88
NFLX 2018-12-21 263.83 246.39
NFLX 2018-12-26 233.92 253.67

Meaning: Lowest closing price records

HANDS-ON 3: ORM TABLE CREATION

Hibernate ORM automatically maps Java classes → database tables.

Entity = Table

Fields = Columns

Primary key = @Id

OUTPUT

employee table created
department table created
skill table created
employee_skill table created

Meaning: Tables created using JPA/Hibernate

HANDS-ON 4: MANY TO ONE

Many employees belong to one department.

Foreign key stored in Employee table

@JoinColumn defines FK

Code

@ManyToOne

@JoinColumn(name = "em_dp_id")

private Department department;

Test

Employee emp = employeeService.get(1);

System.out.println(emp.getDepartment());

OUTPUT

Employee: John

Salary: 50000

Department: IT

Meaning: Employee is linked to department using JOIN

HANDS-ON 5: ONE TO MANY

One department has multiple employees.

mappedBy = reverse side

fetch = EAGER loads data immediately

Code

@OneToMany(mappedBy = "department", fetch = FetchType.EAGER)

private Set<Employee> employeeList;

Test

Department dept = departmentService.get(1);

System.out.println(dept.getEmployeeList());

OUTPUT

Department: IT

Employees:

- John
- Alex
- Ravi

If LAZY (default)

LazyInitializationException

Reason: Data not loaded outside session

HANDS-ON 6: MANY TO MANY

Many employees can have many skills.

Join table used

Extra mapping table created

Code

```
@ManyToMany(fetch = FetchType.EAGER)
@JoinTable(
    name = "employee_skill",
    joinColumns = @JoinColumn(name = "es_em_id"),
    inverseJoinColumns = @JoinColumn(name = "es_sk_id")
)
private Set<Skill> skillList;
```

Test

```
Employee emp = employeeService.get(1);
Skill skill = skillService.get(1);
```

```
emp.getSkillList().add(skill);
employeeService.save(emp);
```

OUTPUT

Employee: John

Department: IT

Skills:

- Java
- Spring Boot
- SQL

DB OUTPUT

employee_skill table updated:

es_em_id | es_sk_id

1 | 2

Many-to-many relationship stored in mapping table

FINAL OUTPUTS

COUNTRY OUTPUT

Bouvet Island

Djibouti

Guadeloupe

Luxembourg

South Sudan

South Africa

Zambia

Zimbabwe

STOCK OUTPUT

FB → September 2019 records

GOOGL → price > 1250

FB → top volume stocks

NFLX → lowest closing prices

ORM OUTPUT

Employee - Department (JOIN result)

Department - Employee List (EAGER fetch)

Employee - Skills (ManyToMany mapping)